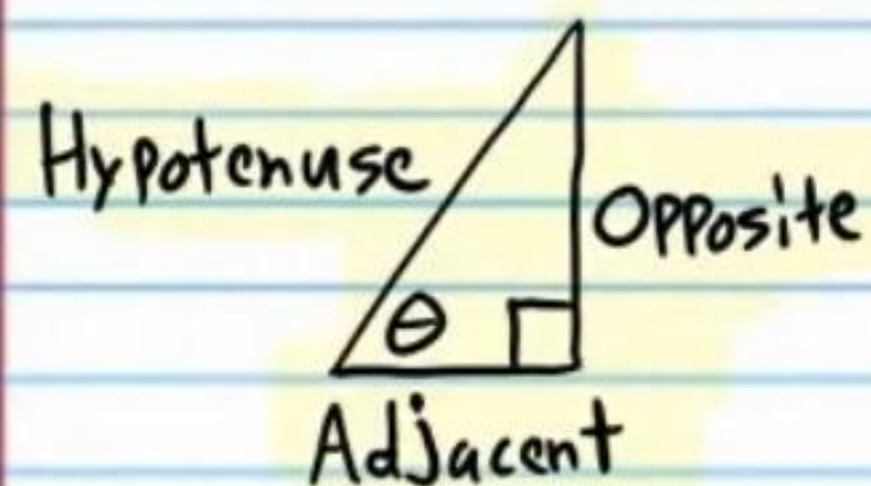


Right Triangles (Part II): The Sine, Cosine, and Tangent Ratios

Trigonometry: The study of relationships between sides and angles of triangles.

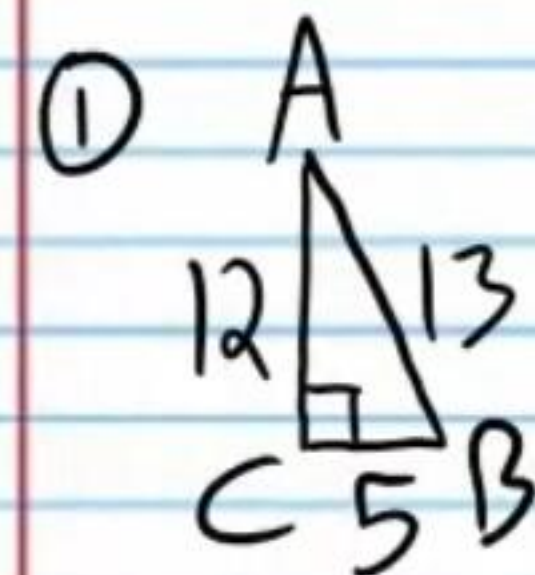


$$\sin \theta = \frac{\text{OPP}}{\text{Hyp}}$$

$$\cos \theta = \frac{\text{Adj}}{\text{Hyp}}$$

$$\tan \theta = \frac{\text{OPP}}{\text{Adj}}$$

For the triangles below, find $\sin A$, $\cos A$, and $\tan A$. Then find $\sin B$, $\cos B$, and $\tan B$.



$$\sin A = \frac{5}{13}$$

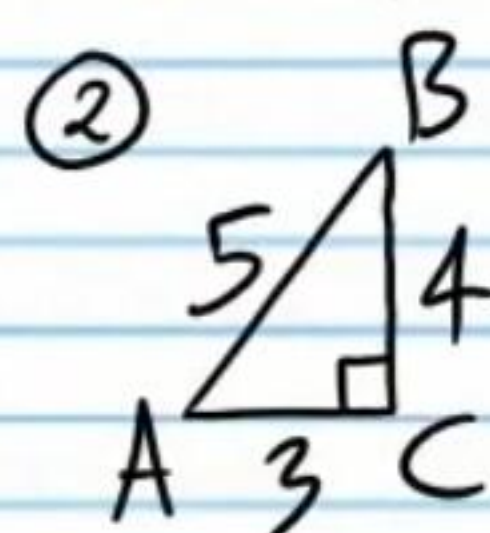
$$\cos A = \frac{12}{13}$$

$$\tan A = \frac{5}{12}$$

$$\sin B = \frac{12}{13}$$

$$\cos B = \frac{5}{13}$$

$$\tan B = \frac{12}{5}$$



$$\sin A = \frac{4}{5}$$

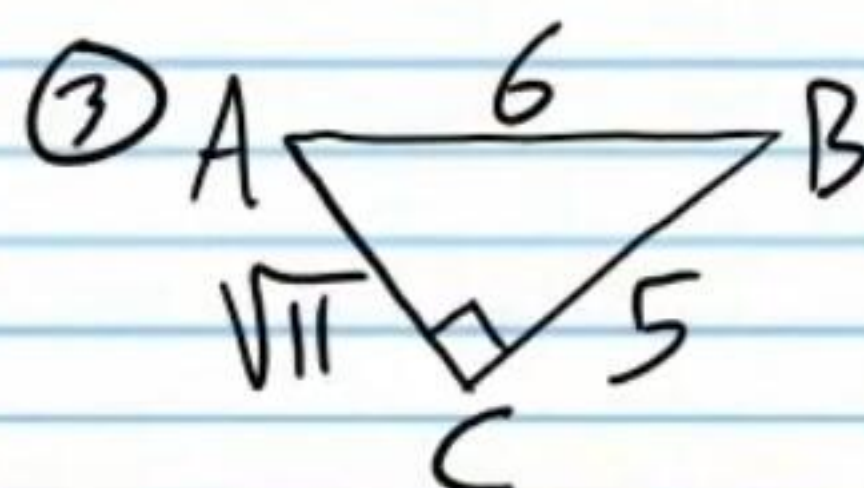
$$\cos A = \frac{3}{5}$$

$$\tan A = \frac{4}{3}$$

$$\sin B = \frac{3}{5}$$

$$\cos B = \frac{4}{5}$$

$$\tan B = \frac{3}{4}$$



$$\sin A = \frac{5}{6}$$

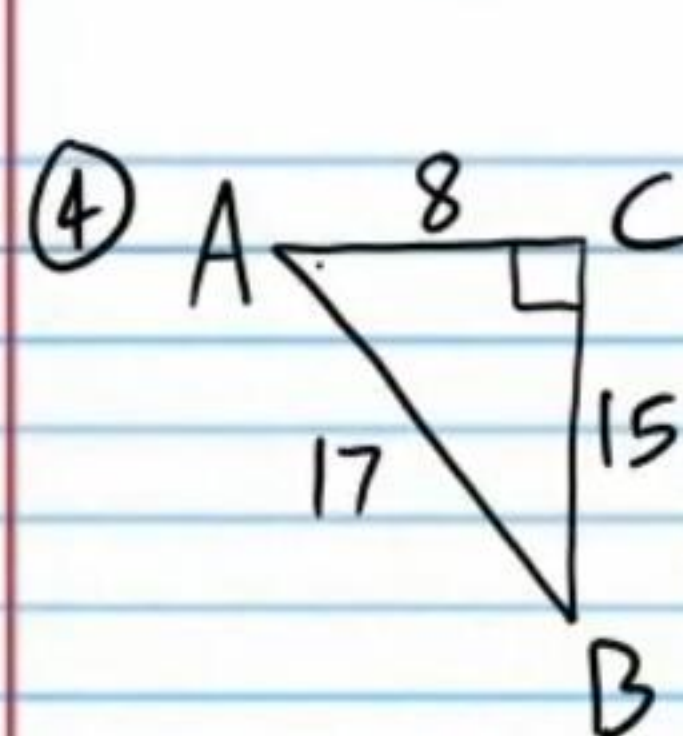
$$\cos A = \frac{\sqrt{11}}{6}$$

$$\tan A = \frac{5}{\sqrt{11}} = \frac{5\sqrt{11}}{11}$$

$$\sin B = \frac{\sqrt{11}}{6}$$

$$\cos B = \frac{5}{6}$$

$$\tan B = \frac{\sqrt{11}}{5}$$



$$\sin A = \frac{15}{17}$$

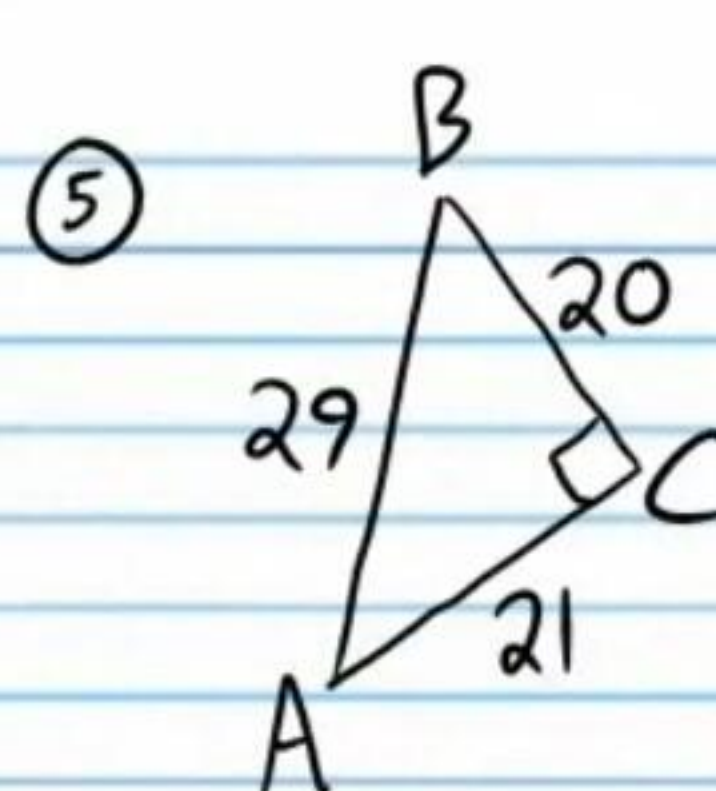
$$\cos A = \frac{8}{17}$$

$$\tan A = \frac{15}{8}$$

$$\sin B = \frac{8}{17}$$

$$\cos B = \frac{15}{17}$$

$$\tan B = \frac{8}{15}$$



$$\sin A = \frac{20}{29}$$

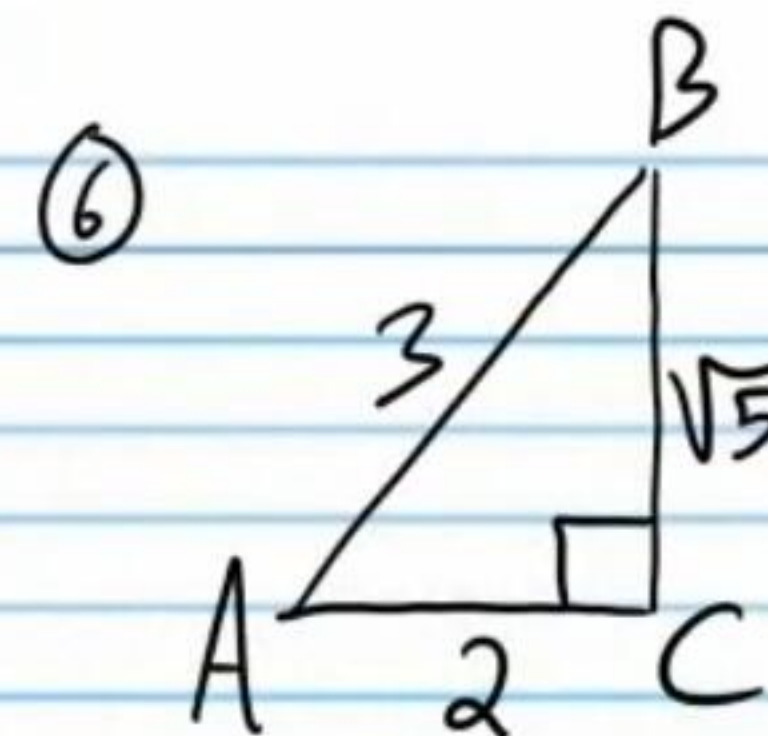
$$\cos A = \frac{21}{29}$$

$$\tan A = \frac{20}{21}$$

$$\sin B = \frac{21}{29}$$

$$\cos B = \frac{20}{29}$$

$$\tan B = \frac{21}{20}$$



$$\sin A = \frac{\sqrt{5}}{3}$$

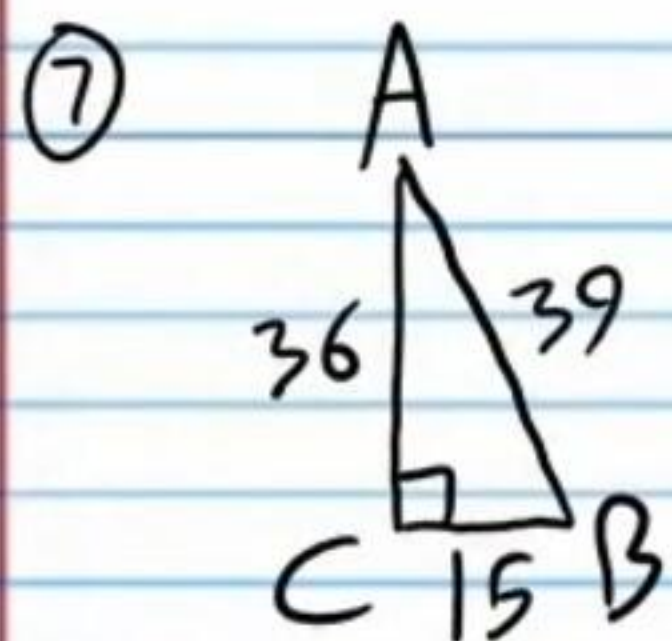
$$\cos A = \frac{2}{3}$$

$$\tan A = \frac{\sqrt{5}}{2}$$

$$\sin B = \frac{2}{3}$$

$$\cos B = \frac{\sqrt{5}}{3}$$

$$\tan B = \frac{2}{\sqrt{5}} = \frac{2\sqrt{5}}{5}$$



$$\sin A = \frac{15}{39} = \boxed{\frac{5}{13}}$$

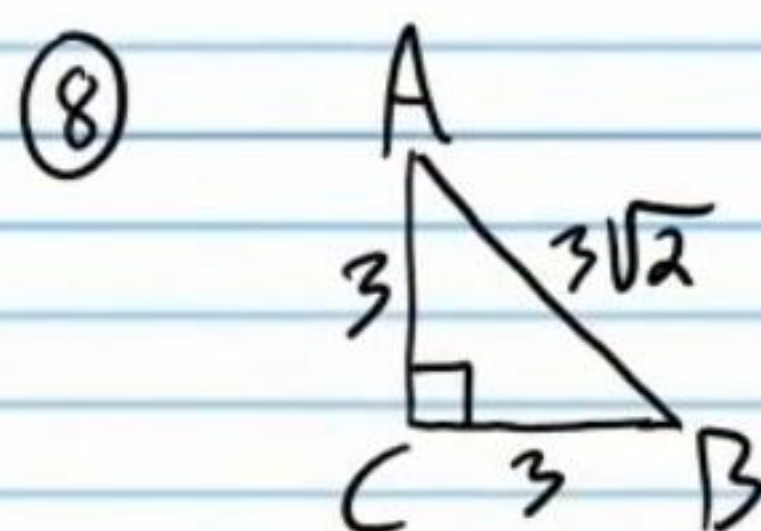
$$\cos A = \frac{36}{39} = \boxed{\frac{12}{13}}$$

$$\tan A = \frac{15}{36} = \boxed{\frac{5}{12}}$$

$$\sin B = \frac{36}{39} = \boxed{\frac{12}{13}}$$

$$\cos B = \frac{15}{39} = \boxed{\frac{5}{13}}$$

$$\tan B = \frac{36}{15} = \boxed{\frac{12}{5}}$$



$$\sin A = \frac{3}{3\sqrt{2}} = \frac{1}{\sqrt{2}} = \boxed{\frac{\sqrt{2}}{2}}$$

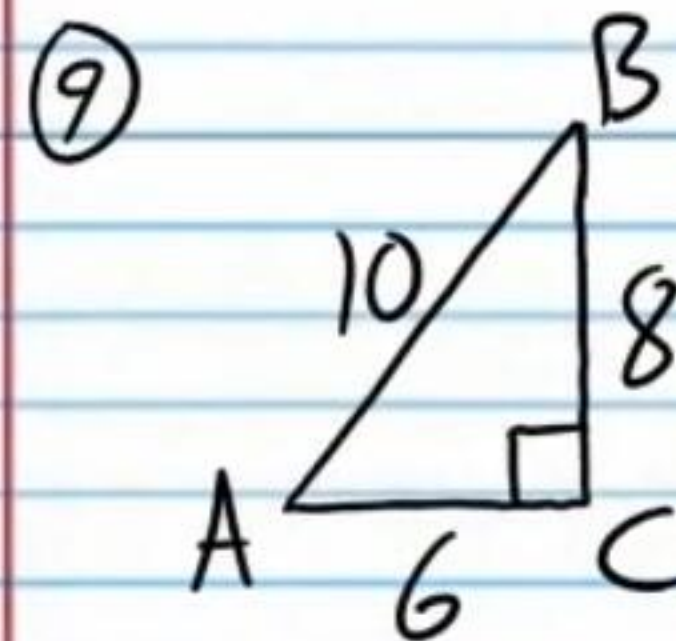
$$\cos A = \frac{3}{3\sqrt{2}} = \frac{1}{\sqrt{2}} = \boxed{\frac{\sqrt{2}}{2}}$$

$$\tan A = \frac{3}{3} = \boxed{1}$$

$$\sin B = \boxed{\frac{\sqrt{2}}{2}}$$

$$\cos B = \boxed{\frac{\sqrt{2}}{2}}$$

$$\tan B = \frac{3}{3} = \boxed{1}$$



$$\sin A = \frac{8}{10} = \boxed{\frac{4}{5}}$$

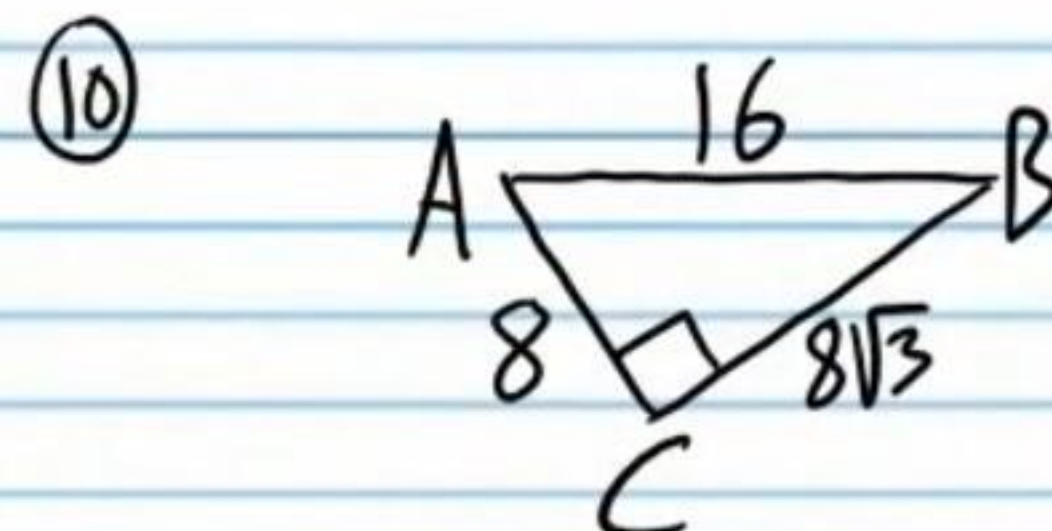
$$\cos A = \frac{6}{10} = \boxed{\frac{3}{5}}$$

$$\tan A = \frac{8}{6} = \boxed{\frac{4}{3}}$$

$$\sin B = \frac{6}{10} = \boxed{\frac{3}{5}}$$

$$\cos B = \frac{8}{10} = \boxed{\frac{4}{5}}$$

$$\tan B = \frac{6}{8} = \boxed{\frac{3}{4}}$$



$$\sin A = \frac{8\sqrt{3}}{16} = \boxed{\frac{\sqrt{3}}{2}}$$

$$\cos A = \frac{8}{16} = \boxed{\frac{1}{2}}$$

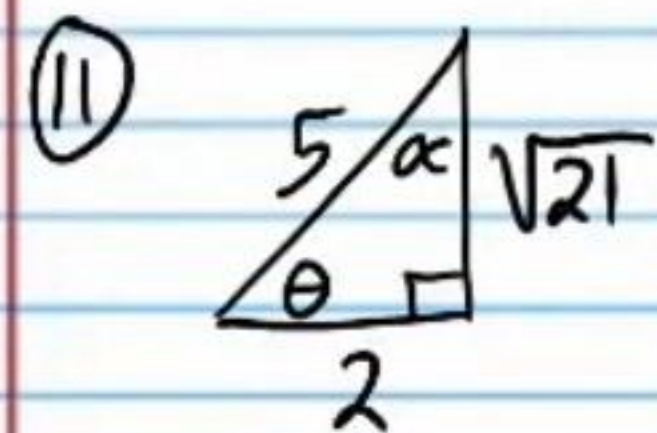
$$\tan A = \frac{8\sqrt{3}}{8} = \boxed{\sqrt{3}}$$

$$\sin B = \frac{8}{16} = \boxed{\frac{1}{2}}$$

$$\cos B = \frac{8\sqrt{3}}{16} = \boxed{\frac{\sqrt{3}}{2}}$$

$$\tan B = \frac{8}{8\sqrt{3}} = \frac{1}{\sqrt{3}} = \boxed{\frac{\sqrt{3}}{3}}$$

Find $\sin \theta$, $\cos \theta$, and $\tan \theta$. Then find $\sin \alpha$, $\cos \alpha$, and $\tan \alpha$.



$$\sin \theta = \frac{\sqrt{21}}{5}$$

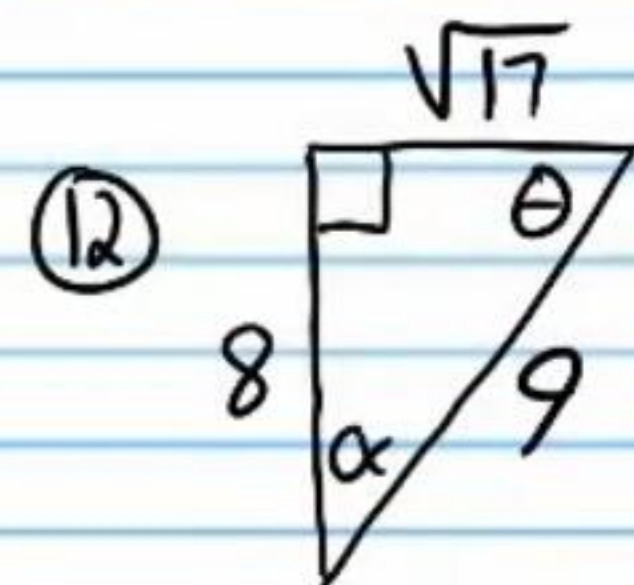
$$\cos \theta = \frac{2}{5}$$

$$\tan \theta = \frac{\sqrt{21}}{2}$$

$$\sin \alpha = \frac{2}{5}$$

$$\cos \alpha = \frac{\sqrt{21}}{5}$$

$$\tan \alpha = \frac{2}{\sqrt{21}} = \frac{2\sqrt{21}}{21}$$



$$\sin \theta = \frac{8}{9}$$

$$\cos \theta = \frac{\sqrt{17}}{9}$$

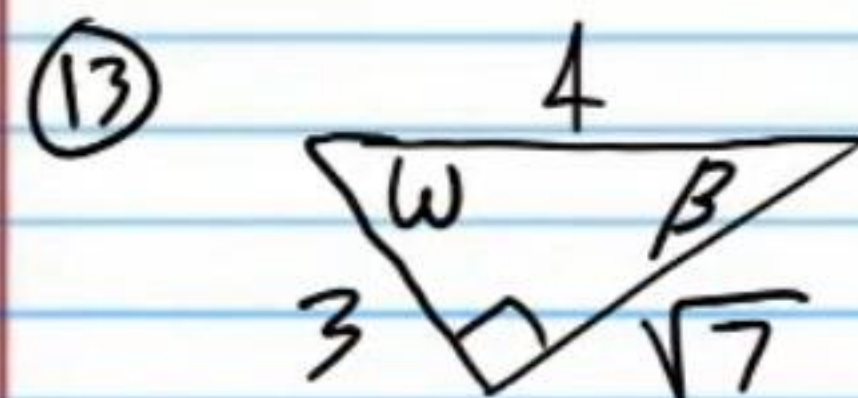
$$\tan \theta = \frac{8}{\sqrt{17}} = \frac{8\sqrt{17}}{17}$$

$$\sin \alpha = \frac{\sqrt{17}}{9}$$

$$\cos \alpha = \frac{8}{9}$$

$$\tan \alpha = \frac{\sqrt{17}}{8}$$

Find $\sin w$, $\cos w$, and $\tan w$. Then find $\sin \beta$, $\cos \beta$, and $\tan \beta$.



$$\sin w = \frac{\sqrt{7}}{4}$$

$$\cos w = \frac{3}{4}$$

$$\tan w = \frac{\sqrt{7}}{3}$$

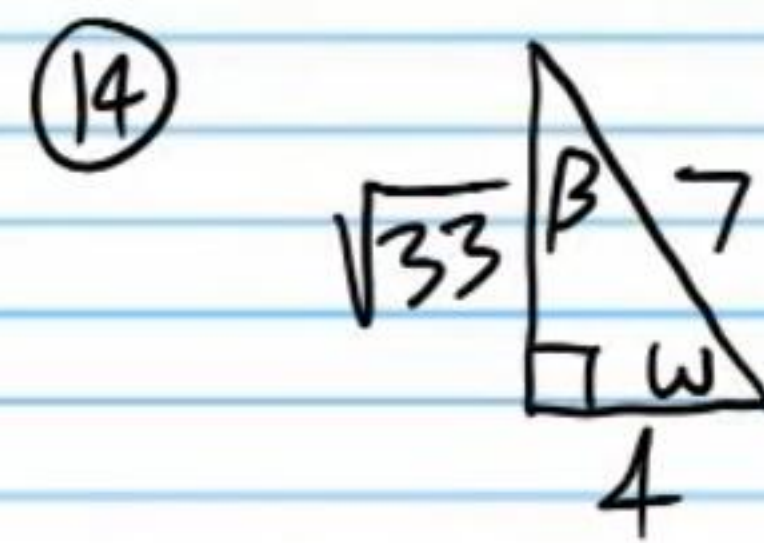
$$\sin \beta = \frac{3}{4}$$

$$\cos \beta = \frac{\sqrt{7}}{4}$$

$$\tan \beta = \frac{3}{\sqrt{7}} = \frac{3 \cdot \sqrt{7}}{\sqrt{7} \cdot \sqrt{7}}$$

$$= \frac{3\sqrt{7}}{\sqrt{49}}$$

$$= \frac{3\sqrt{7}}{7}$$



$$\sin w = \frac{\sqrt{33}}{7}$$

$$\cos w = \frac{4}{7}$$

$$\tan w = \frac{\sqrt{33}}{4}$$

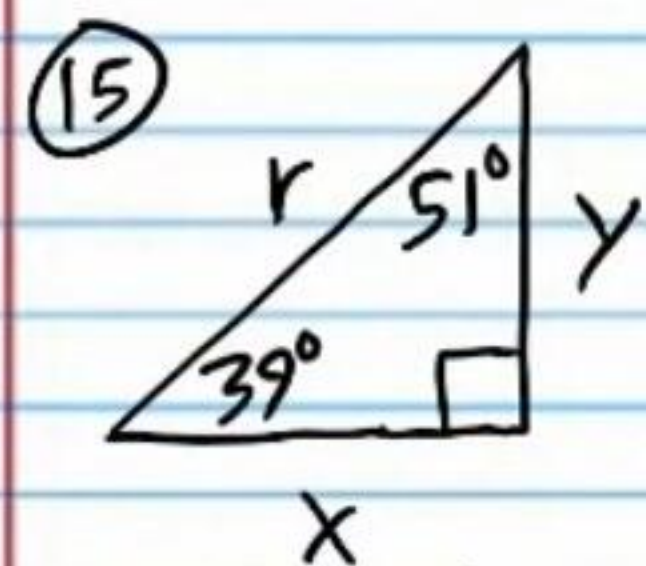
$$\sin \beta = \frac{4}{7}$$

$$\cos \beta = \frac{\sqrt{33}}{7}$$

$$\tan \beta = \frac{4}{\sqrt{33}} = \frac{4 \cdot \sqrt{33}}{\sqrt{33} \cdot \sqrt{33}}$$

$$= \frac{4\sqrt{33}}{33}$$

Find the ratios listed below.



$$\sin 39^\circ = \frac{y}{r}$$

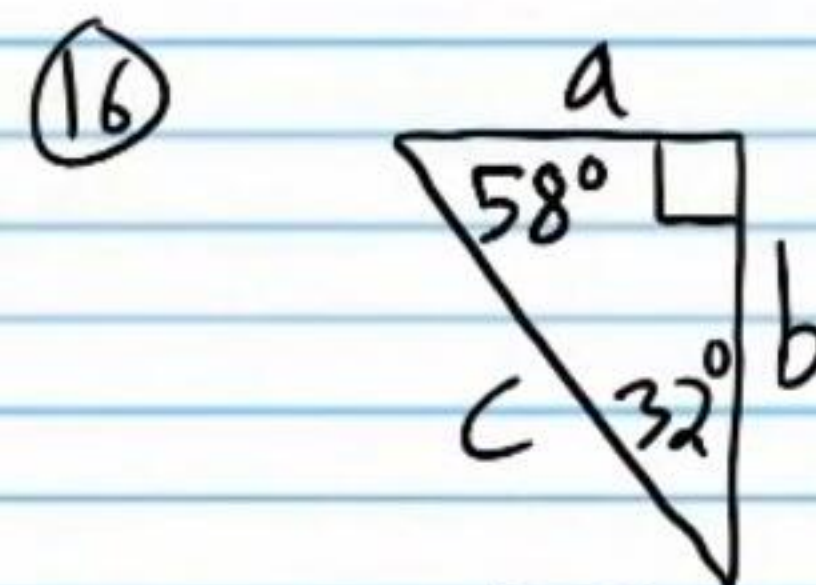
$$\cos 39^\circ = \frac{x}{r}$$

$$\tan 39^\circ = \frac{y}{x}$$

$$\sin 51^\circ = \frac{x}{r}$$

$$\cos 51^\circ = \frac{y}{r}$$

$$\tan 51^\circ = \frac{x}{y}$$



$$\sin 58^\circ = \frac{b}{c}$$

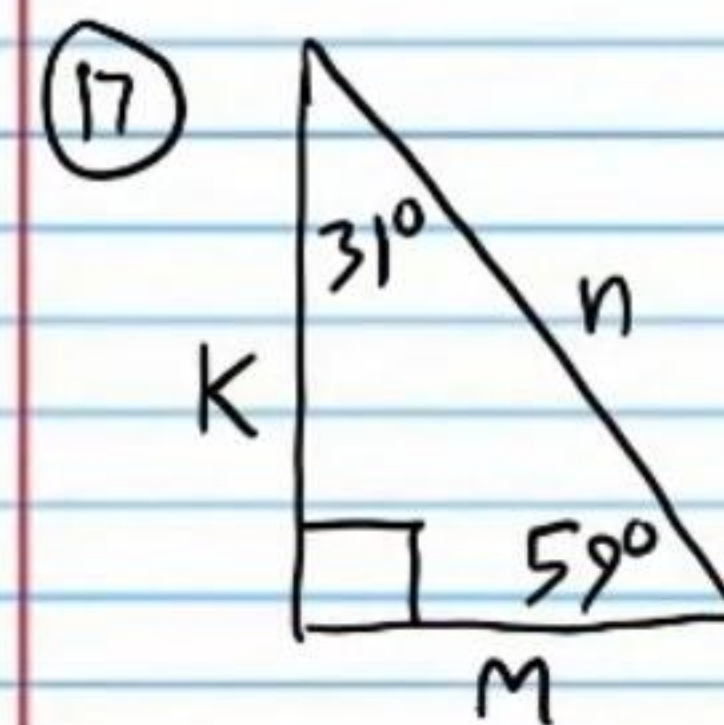
$$\cos 58^\circ = \frac{a}{c}$$

$$\tan 58^\circ = \frac{b}{a}$$

$$\sin 32^\circ = \frac{a}{c}$$

$$\cos 32^\circ = \frac{b}{c}$$

$$\tan 32^\circ = \frac{a}{b}$$



$$\sin 59^\circ = \frac{k}{n}$$

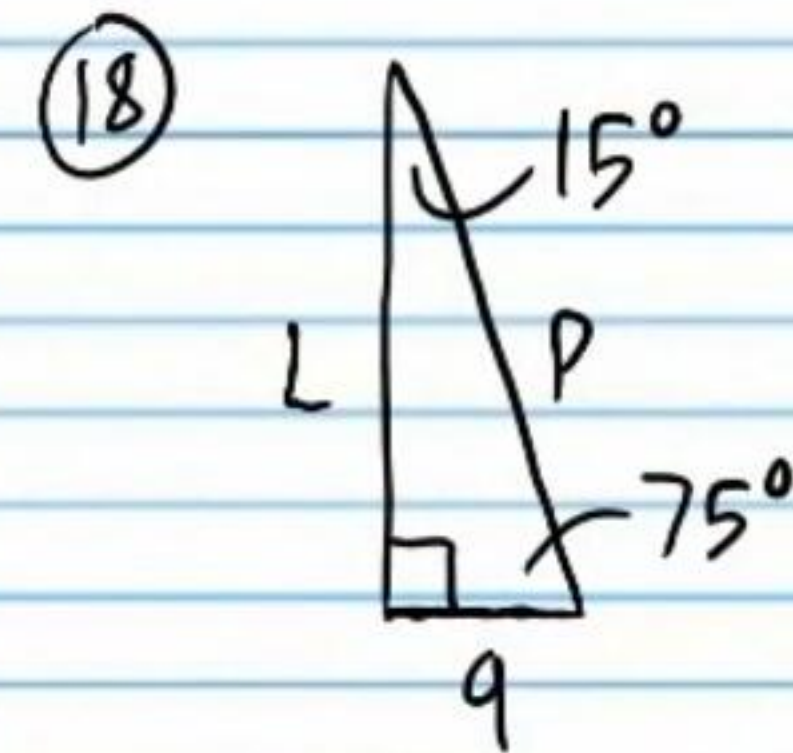
$$\cos 59^\circ = \frac{m}{n}$$

$$\tan 59^\circ = \frac{k}{m}$$

$$\sin 31^\circ = \frac{m}{n}$$

$$\cos 31^\circ = \frac{k}{n}$$

$$\tan 31^\circ = \frac{m}{k}$$



$$\sin 15^\circ = \frac{q}{l}$$

$$\cos 15^\circ = \frac{p}{l}$$

$$\tan 15^\circ = \frac{q}{p}$$

$$\sin 75^\circ = \frac{p}{l}$$

$$\cos 75^\circ = \frac{q}{l}$$

$$\tan 75^\circ = \frac{p}{q}$$

Finding Ratios with a Calculator

Use a calculator to find the sine, cosine, and tangent of each of the following angles. Round to four decimal places.

★ Be sure that your calculator is in degree mode, not radian mode.

(19) $\theta = 82^\circ$

$$\sin 82^\circ \approx 0.9902680687$$

$$\approx \boxed{0.9903}$$

$$\cos 82^\circ \approx 0.13917310096$$

$$\approx \boxed{0.1392}$$

$$\tan 82^\circ \approx 7.115369722$$

$$\approx \boxed{7.1154}$$

(21) $\theta = 49^\circ$

$$\sin 49^\circ \approx 0.754709$$

$$\approx \boxed{0.7547}$$

$$\cos 49^\circ \approx 0.656059$$

$$\tan 49^\circ \approx \boxed{0.6561}$$

$$\approx \boxed{1.1504}$$

(20) $\theta = 25^\circ$

$$\sin 25^\circ \approx 0.422618$$

$$\approx \boxed{0.4226}$$

$$\cos 25^\circ \approx 0.9063077$$

$$\approx \boxed{0.9063}$$

$$\tan 25^\circ \approx 0.466307$$

$$\approx \boxed{0.4663}$$

(22) $\theta = 17^\circ$

$$\sin 17^\circ \approx \boxed{0.2924}$$

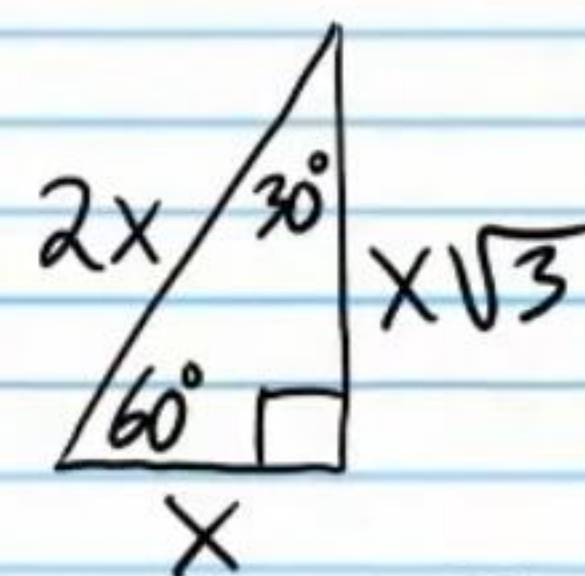
$$\cos 17^\circ \approx \boxed{0.9563}$$

$$\tan 17^\circ \approx \boxed{0.3057}$$

Finding Special Ratios without a Calculator

Find the ratios listed below.

(23)

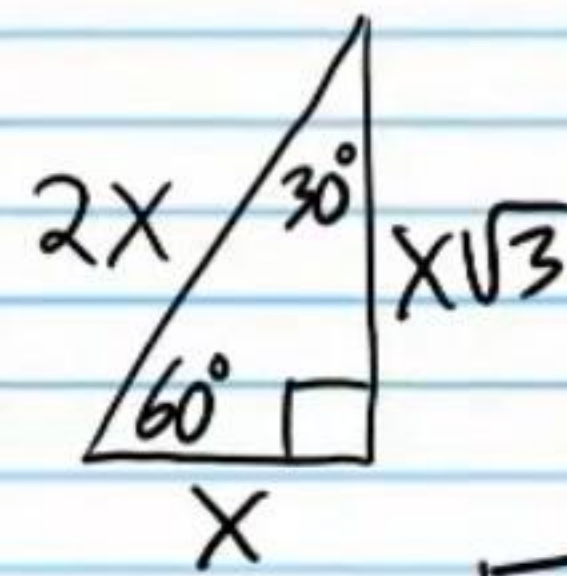


$$\sin 60^\circ = \frac{x\sqrt{3}}{2x} = \boxed{\frac{\sqrt{3}}{2}}$$

$$\cos 60^\circ = \frac{x}{2x} = \boxed{\frac{1}{2}}$$

$$\tan 30^\circ = \frac{x}{x\sqrt{3}} = \frac{1}{\sqrt{3}} = \boxed{\frac{\sqrt{3}}{3}}$$

(24)

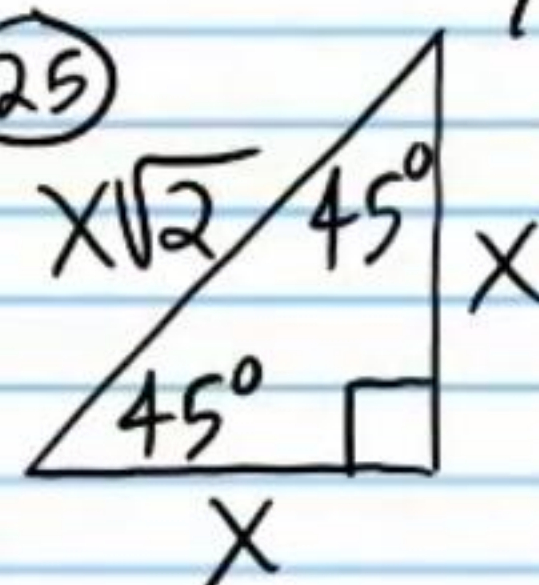


$$\sin 30^\circ = \frac{x}{2x} = \boxed{\frac{1}{2}}$$

$$\cos 30^\circ = \frac{x\sqrt{3}}{2x} = \boxed{\frac{\sqrt{3}}{2}}$$

$$\tan 60^\circ = \frac{x\sqrt{3}}{x} = \boxed{\sqrt{3}}$$

(25)

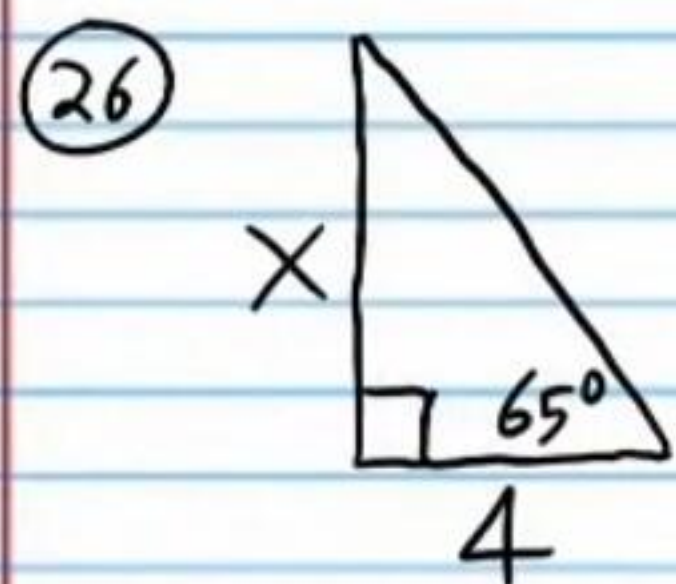


$$\sin 45^\circ = \frac{x}{x\sqrt{2}} = \frac{1}{\sqrt{2}} = \boxed{\frac{\sqrt{2}}{2}}$$

$$\cos 45^\circ = \frac{x}{x\sqrt{2}} = \frac{1}{\sqrt{2}} = \boxed{\frac{\sqrt{2}}{2}}$$

$$\tan 45^\circ = \frac{x}{x} = \boxed{1}$$

Find X. Round to four decimal places.

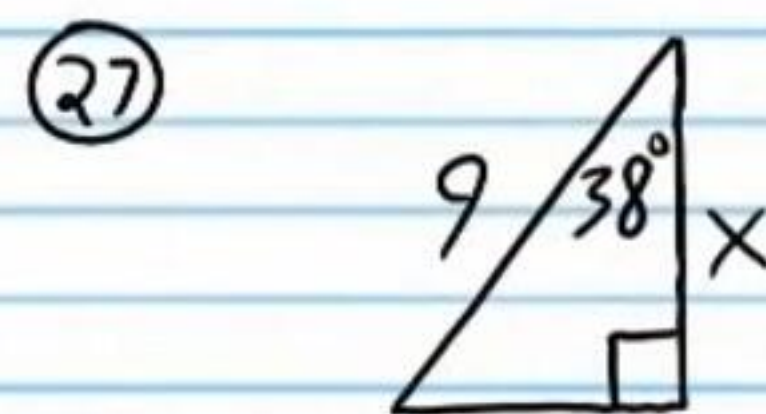


$$\tan 65^\circ = \frac{X}{4}$$

$$(4) \tan 65^\circ = \frac{X}{4} (4)$$

$$4 \tan 65^\circ = X$$

$$\boxed{8.5780 \approx X}$$

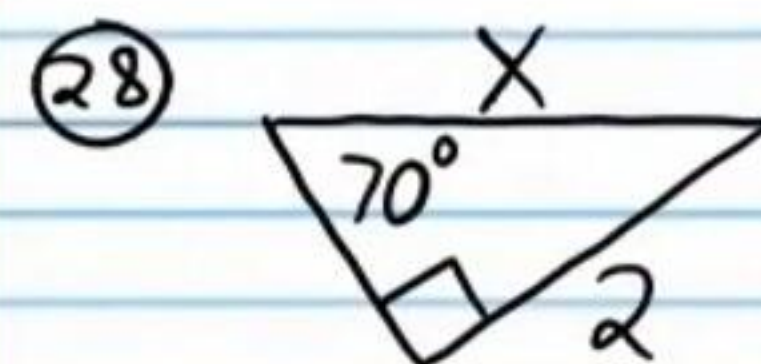


$$\cos 38^\circ = \frac{X}{9}$$

$$(9) \cos 38^\circ = \frac{X}{9} (9)$$

$$9 \cos 38^\circ = X$$

$$\boxed{7.0921 \approx X}$$



$$\sin 70^\circ = \frac{2}{X}$$

$$\frac{\sin 70^\circ}{1} = \frac{2}{X}$$

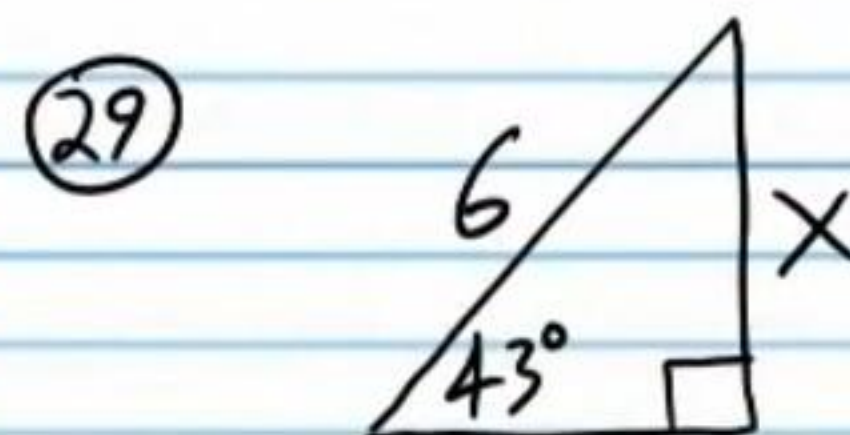
~~$$\frac{\sin 70^\circ}{1} = \frac{2}{X}$$~~

$$X \sin 70^\circ = 2$$

$$\frac{X \sin 70^\circ}{\sin 70^\circ} = \frac{2}{\sin 70^\circ}$$

$$X = \frac{2}{\sin 70^\circ}$$

$$\boxed{X \approx 2.1284}$$

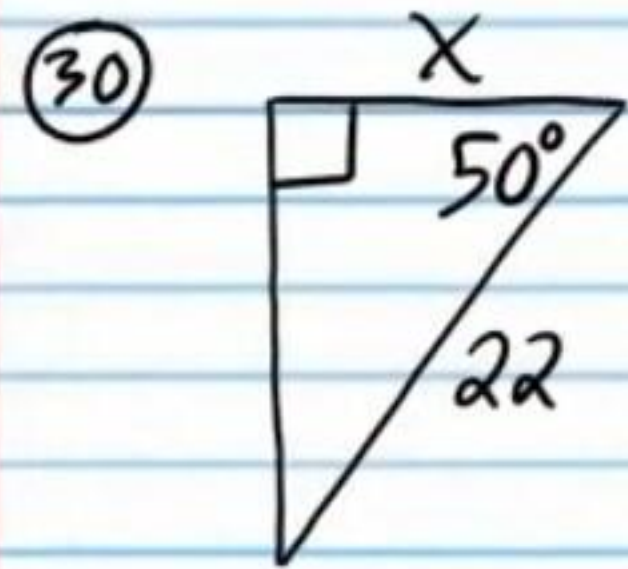


$$\sin 43^\circ = \frac{X}{6}$$

$$(6) \sin 43^\circ = \frac{X}{6} (6)$$

$$6 \sin 43^\circ = X$$

$$\boxed{4.092 \approx X}$$

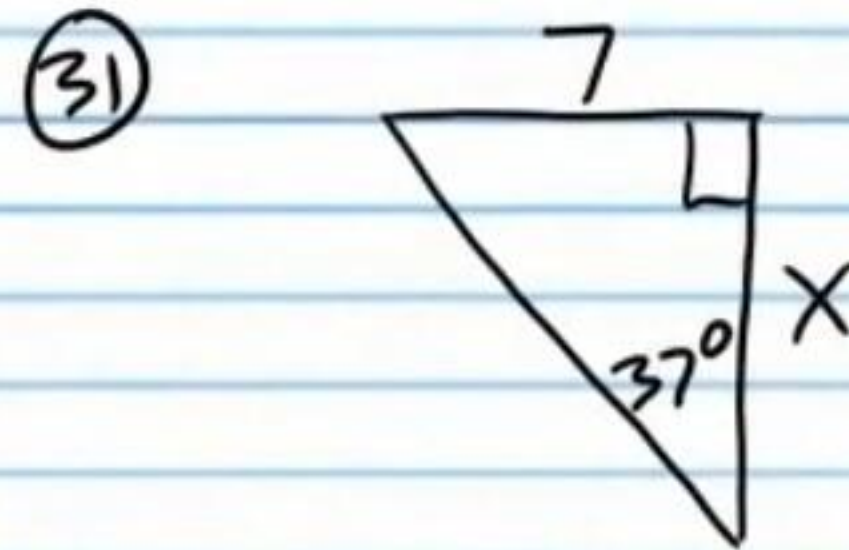


$$\cos 50^\circ = \frac{x}{22}$$

$$(22) \cos 50^\circ = \frac{x}{22} (22)$$

$$22 \cos 50^\circ = x$$

$$\boxed{14.1413 \approx x}$$



$$\tan 37^\circ = \frac{7}{x}$$

$$\frac{\tan 37^\circ}{1} = \frac{7}{x}$$

~~$$\frac{\tan 37^\circ}{1} = \frac{7}{x}$$~~

$$7 = x \tan 37^\circ$$

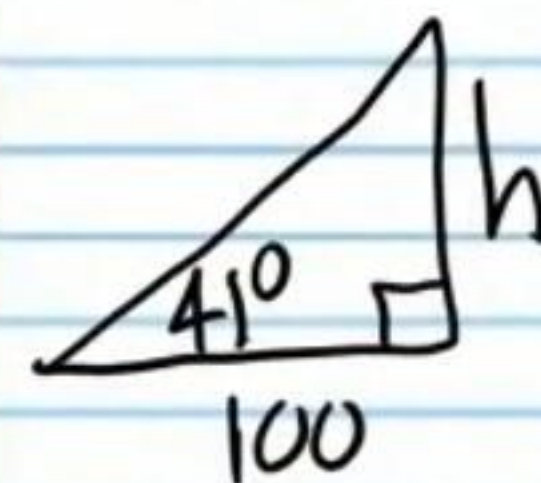
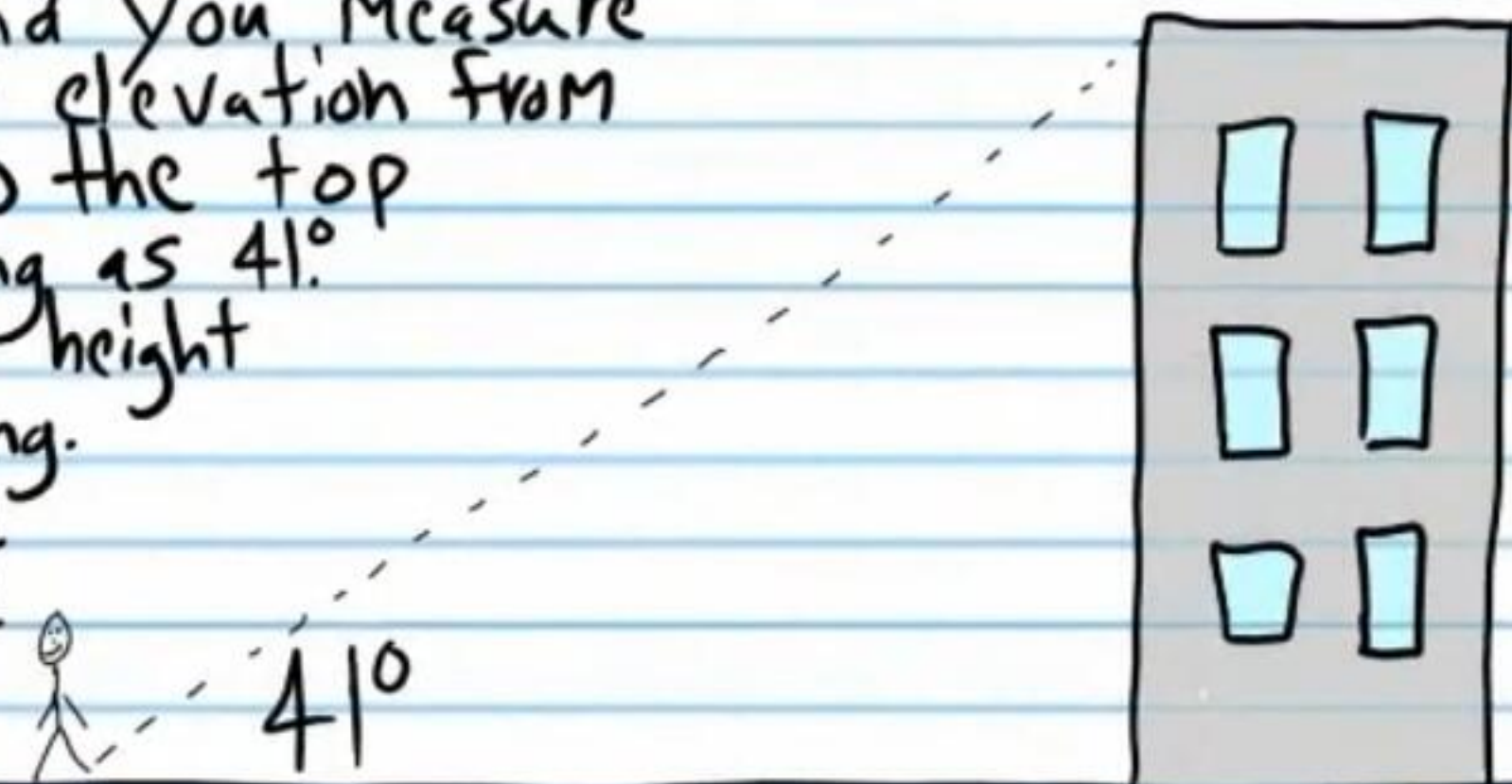
$$\frac{7}{\tan 37^\circ} = \frac{x \tan 37^\circ}{\tan 37^\circ}$$

$$\frac{7}{\tan 37^\circ} = x$$

$$\boxed{9.2893 \approx x}$$

Angle of Elevation and Angle of Depression

(32) You stand 100 feet from a building and you measure the angle of elevation from the ground to the top of the building as 41° . Determine the height of the building. Round to the nearest whole number.



100 ft

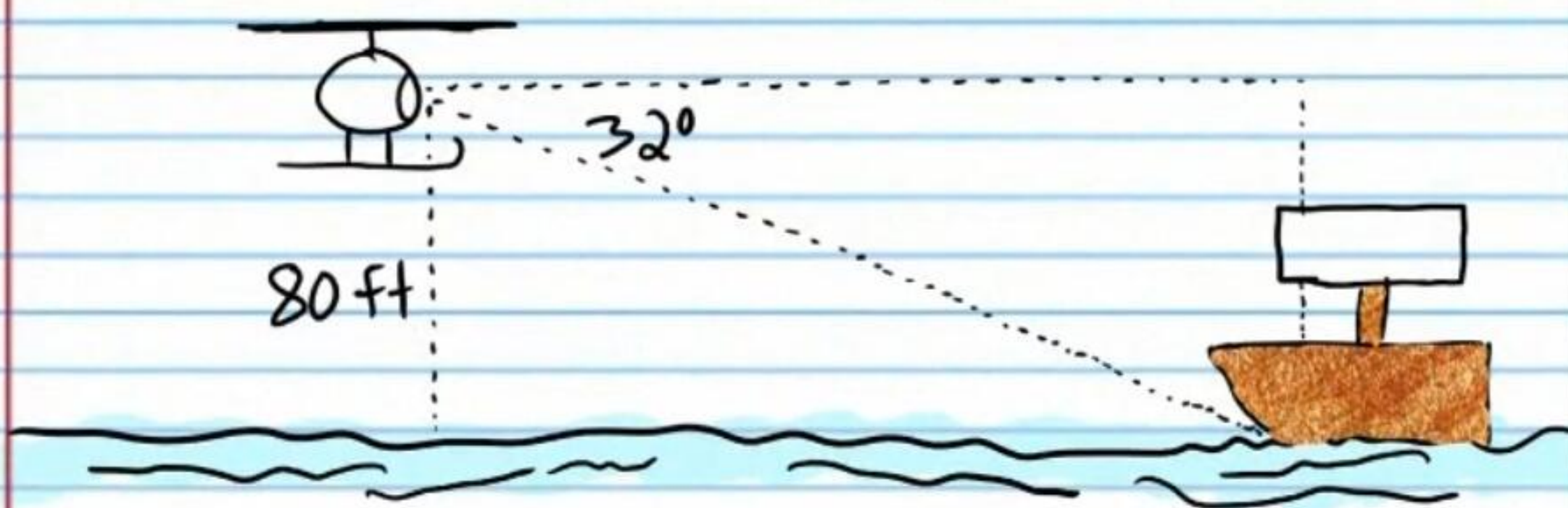
$$\tan 41^\circ = \frac{h}{100}$$

$$(100) \tan 41^\circ = \frac{h}{100} (100)$$

$$100 \tan 41^\circ = h$$

$$\boxed{87 \text{ ft} \approx h}$$

33) The angle of depression from a helicopter to a ship is 32° . If the pilot knows that he is flying 80 feet above the water, what is the horizontal distance from the helicopter to the ship? Round to the nearest whole number.



$$\tan 32^\circ = \frac{80}{X}$$

$$\frac{\tan 32^\circ}{1} = \frac{80}{X}$$

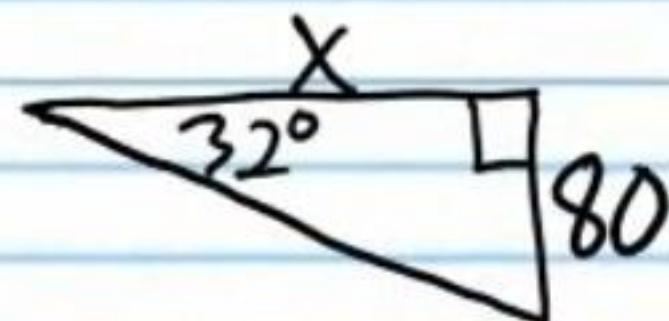
~~$$\frac{\tan 32^\circ}{1} = \frac{80}{X}$$~~

$$X \tan 32^\circ = 80$$

$$\frac{X \tan 32^\circ}{\tan 32^\circ} = \frac{80}{\tan 32^\circ}$$

$$X = \frac{80}{\tan 32^\circ}$$

$$X \approx 128 \text{ ft}$$



34) For problem # 33, what is the length of the shortest path from the helicopter to the ship? Round to the nearest whole number.

$$\sin 32^\circ = \frac{80}{X}$$

$$\frac{\sin 32^\circ}{1} = \frac{80}{X}$$

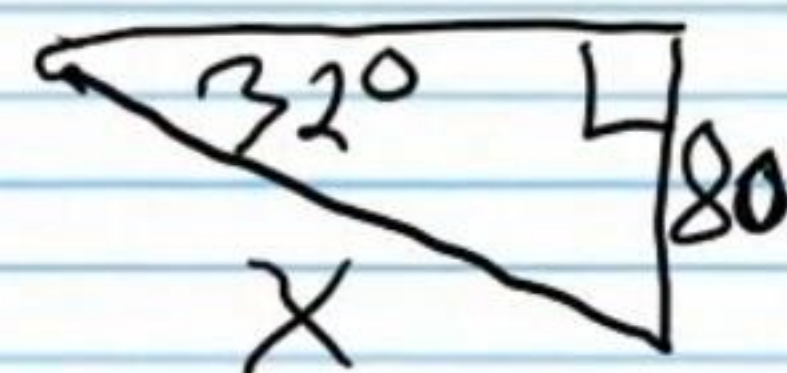
~~$$\frac{\sin 32^\circ}{1} = \frac{80}{X}$$~~

$$X \sin 32^\circ = 80$$

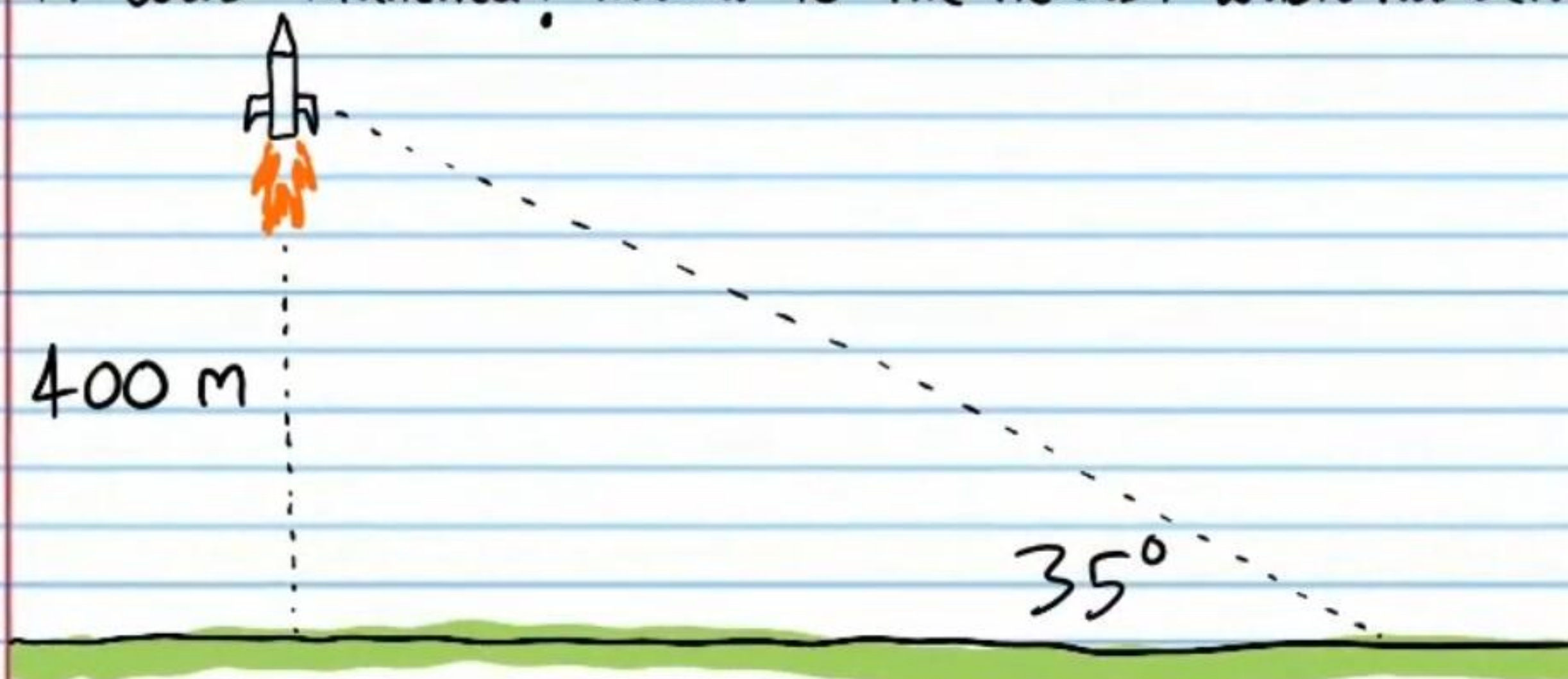
$$\frac{X \sin 32^\circ}{\sin 32^\circ} = \frac{80}{\sin 32^\circ}$$

$$X = \frac{80}{\sin 32^\circ}$$

$$X \approx 151 \text{ ft}$$



35) Some kids launch a toy rocket at the park. After one second in the air, its altitude is 400 meters. If the angle of elevation from the ground where the children are sitting is 35° , how far did the kids move from the rocket before it was launched? Round to the nearest whole number.



$$\tan 35^\circ = \frac{400}{x}$$

$$\frac{\tan 35^\circ}{1} = \frac{400}{x}$$

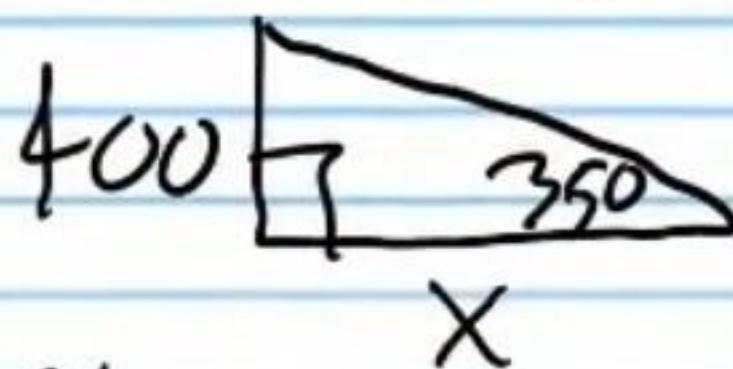
~~$$\frac{\tan 35^\circ}{1} = \frac{400}{x}$$~~

$$x \tan 35^\circ = 400$$

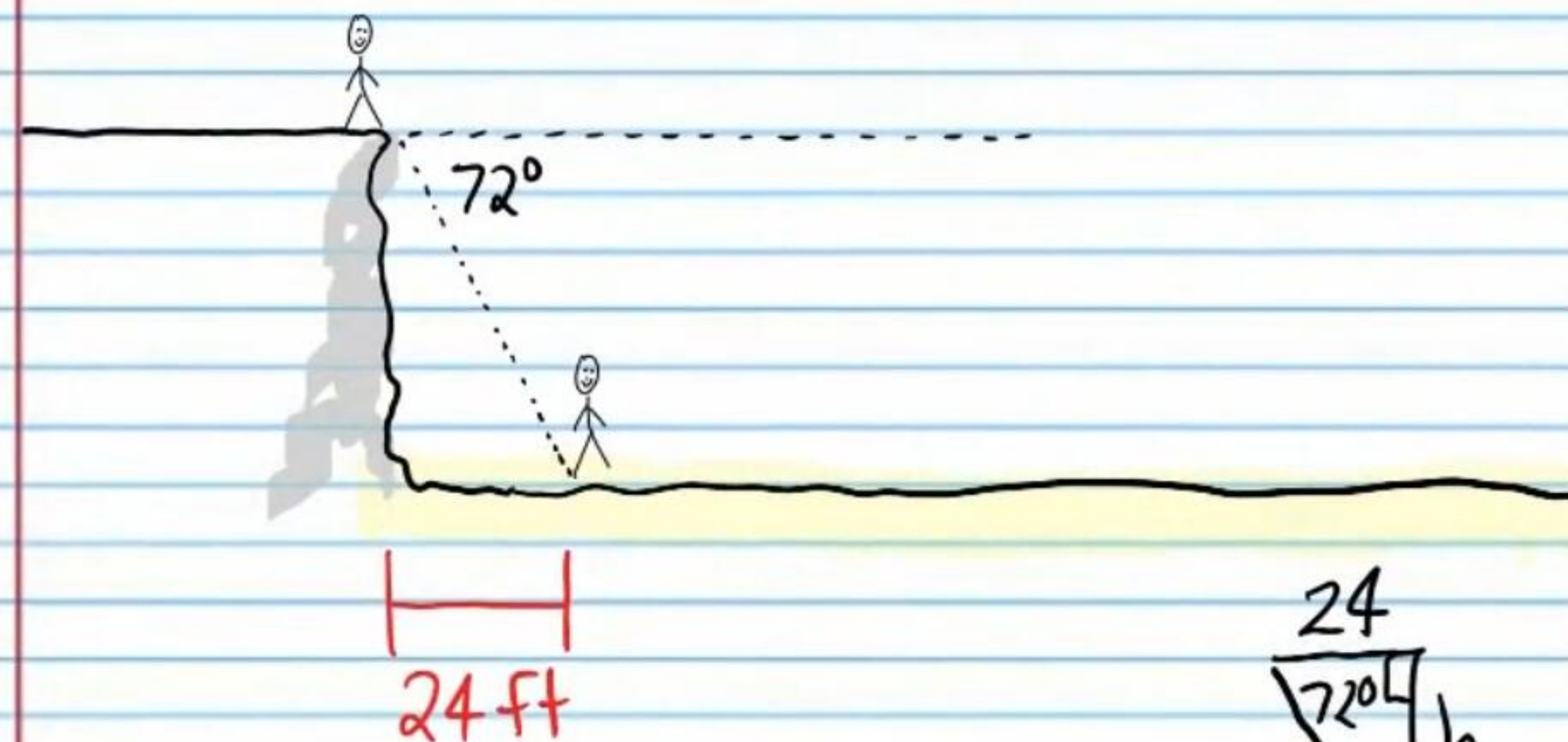
$$\frac{x \tan 35^\circ}{\tan 35^\circ} = \frac{400}{\tan 35^\circ}$$

$$x = \frac{400}{\tan 35^\circ}$$

$$\boxed{x \approx 571 \text{ m}}$$



36) Steve is on the edge of a cliff looking down to Joe. The angle of depression from the edge of the cliff to Joe is 72° . If Joe is 24 feet from the base of the cliff, how high is the cliff? Round to the nearest whole number.

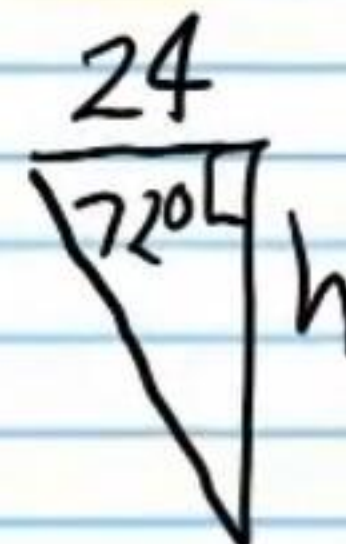


$$\tan 72^\circ = \frac{h}{24}$$

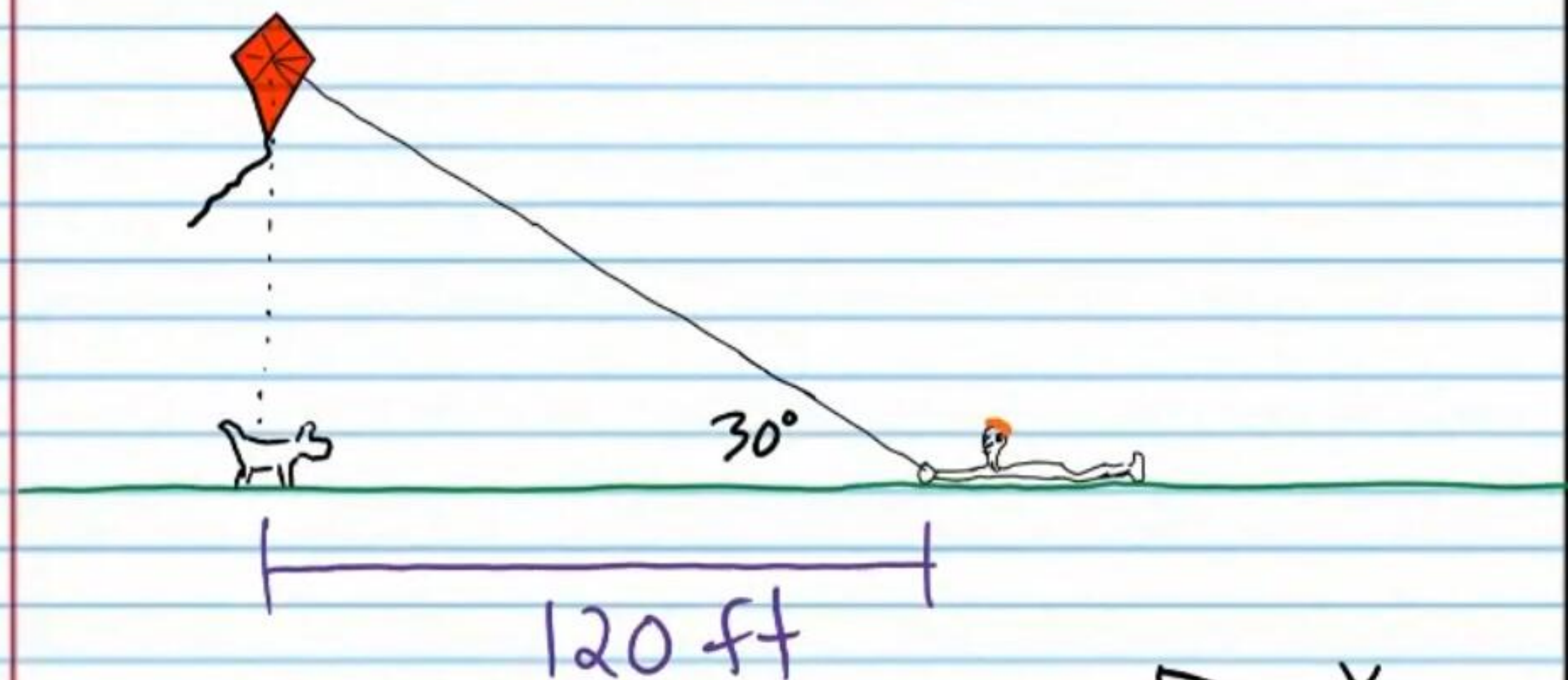
$$(24) \tan 72^\circ = \frac{h}{24} (24)$$

$$24 \tan 72^\circ = h$$

$$\boxed{74 \text{ ft} \approx h}$$



37) Ryan's kite is flying directly above his dog Coco. If Coco is 120 feet from Ryan, what is the length of the string that Ryan let out to raise the kite? The angle of elevation from Ryan's hands to the kite is 30° . Round to the nearest whole number.

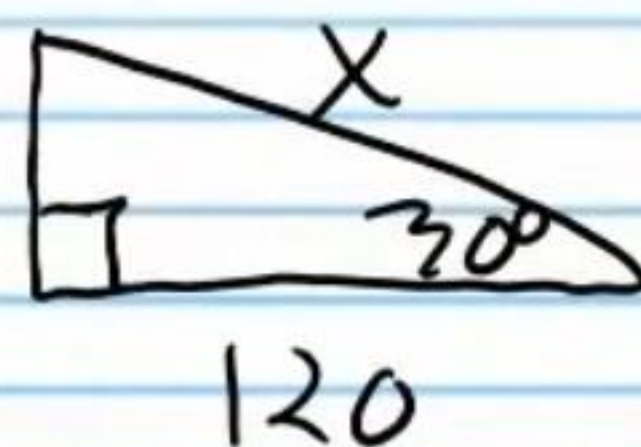


$$\cos 30^\circ = \frac{120}{x}$$

$$\frac{\cos 30^\circ}{1} = \frac{120}{x}$$

~~$$\frac{\cos 30^\circ}{1} = \frac{120}{x}$$~~

$$x \cos 30^\circ = 120$$



$$\frac{x \cos 30^\circ}{\cos 30^\circ} = \frac{120}{\cos 30^\circ}$$

$$x = \frac{120}{\cos 30^\circ}$$

$$x \approx 139 \text{ ft}$$

